



GPU FinOps & Cost Optimization

AICB-P2T2 · Ngày 25 · Chương 5: Vận Hành

Giảng viên

VinUniversity · Phase 2 · Track 2 · Tuần 5

HÃY SUY NGHĨ...

*“Bạn đang tiêu bao nhiêu cho GPU mỗi ngày?
Và bao nhiêu % là lãng phí? **Case study:**
4x A100 idle overnight (12h) = \$144 wasted/day = \$52,560/year. Hôm nay chúng ta học cách cắt giảm 40–60% chi phí GPU — và close Chương 5 với Quiz + Milestone 2.”*

Giữ câu hỏi này trong đầu khi học bài hôm nay

1. GPU Cloud Cost Anatomy
2. Spot & Preemptible Instances
3. Right-Sizing & Utilization
4. Inference Cost Optimization
5. Cost Allocation & Chargeback
6. Sustainable AI: Carbon & Energy
7. Tổng Kết Chương 5
8. **Quiz + Milestone 2**

Sau buổi học này, bạn sẽ:

1. Phân tích chi phí GPU cloud và phát hiện lãng phí
2. Áp dụng spot/preemptible instances với checkpoint strategy
3. Tối ưu training cost bằng mixed precision (AMP) và autoscaling
4. Tối ưu inference cost bằng batching, caching, model cascading
5. Thiết kế cost allocation & FinOps review process cho team

Cost anatomy → Spot strategy → Right-sizing + Autoscaling → Mixed precision → Inference optimization → FinOps governance → Quiz + Milestone 2

Lab notebook hoàn thành (8 Parts) + cost charts + Quiz + Milestone 2 demo

- **Parts 1–5:** Mock cluster monitoring, spot savings, autoscaler, waste report
- **Parts 6–7:** Visualization charts + full FinOps workflow end-to-end
- **Part 8:** Real GPU training FP32 vs AMP — time/memory/cost comparison
- **Quiz Chương 5:** 15 câu hỏi CI/CD, LLMOps, Monitoring, Governance, FinOps
- **Milestone 2:** Demo operations platform end-to-end



- **Hidden costs:** data transfer egress (\$0.09/GB AWS), NAT gateway (\$0.045/GB), Secrets Manager (\$0.40/secret/mo)
- **Wasted spend:** idle GPUs (training done, instance running), over-provisioned instances, unused reserved capacity
- **FinOps maturity:** Inform → Optimize → Operate — hầu hết AI teams chỉ ở Inform level

1. GPU idle overnight — training xong nhưng instance vẫn chạy
2. Over-provisioned: dùng H100 cho inference 8B model (A10G đủ rồi)
3. Unused reserved capacity mua 1 năm nhưng workload thay đổi
4. Development environments chạy 24/7 (chỉ cần giờ hành chính)

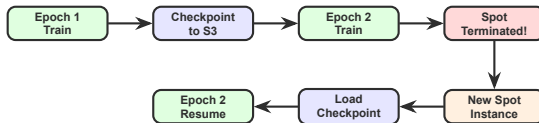
- 4x A100 idle 12h/day
- Chi phí: $4 \times \$3.0/\text{hr} \times 12\text{h} = \$144/\text{ngày}$
- **\$52,560/năm** — chỉ riêng idle time!
- Fix: auto-shutdown schedule → tiết kiệm ngay

Rule of thumb: GPU utilization $< 30\%$ = cần right-size ngay

- Discount 60–70% so với on-demand
- 2-min termination notice
- Spot Fleet: request từ nhiều AZ/instance types
- Giảm interruption rate từ 15% → 3%

- Discount tới 80%
- Terminate sau 24h (Preemptible) hoặc flexible (Spot)
- Phù hợp jobs <20h
- Tự động reschedule trên GKE

Mixed fleet strategy: 20% on-demand (baseline) + 80% spot (burst) — balance cost vs reliability **SkyPilot:** Multi-cloud spot abstraction layer — tự động tìm cheapest spot across AWS/GCP/Azure. `sky launch task.yaml` chọn provider tối ưu.



- Save model state mỗi epoch (hoặc mỗi 30 phút cho long epochs)
- Checkpoint lưu lên S3/GCS — resume từ bất kỳ instance nào
- PyTorch Lightning `ModelCheckpoint` callback tự động hoá
- Best practice: test resume flow **trước** khi chạy long training

Workload	Typical Util	Target	Action nếu thấp
Inference (single model)	20–40%	>60%	Multi-model serving, MIG
Inference (batched)	50–70%	>75%	Tune batch size, queue
Fine-tuning	60–80%	>80%	Larger batch, gradient accum
Pre-training	80–95%	>90%	Optimise data loading

Monitoring: `nvidia-smi dmon -d 5` hoặc DCGM Exporter → Prometheus → Grafana dashboard

- vLLM serve Llama-3-8B + Mistral-7B trên A10G (24GB)
- Dynamic loading: swap models theo request
- Utilization tăng từ 25% → 65%

- A100 80GB → 7 isolated instances (3g.20gb)
- 7 models chạy song song, isolation đảm bảo
- K8s: `nvidia.com/gpu.shared: true`
- Perfect cho inference farm

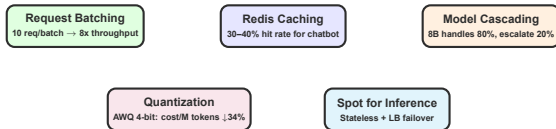
Vertical Pod Autoscaler (VPA): recommend CPU/memory limits dựa trên actual usage — avoid over-provisioning resources.

- **Scale-up:** GPU utilization $>80\%$ → add node
- **Scale-down:** utilization $<20\%$ → remove idle node
- **Cooldown:** 60s giữa các scaling events
- **Bounds:** min 1 node, max 8 nodes
- KEDA: event-driven autoscaling cho K8s

- Prefer cheapest GPU type khi scale-up (T4 trước A100)
- Scale-down idle nodes trước — tiết kiệm ngay
- Spot instances cho burst capacity
- Monitor: nếu idle $>50\%$ GPUs → scale down

Key metric: Cost per useful GPU-hour, không phải tổng spend

Lab demo: Configure autoscaler policy → submit workloads → observe scale-up/down decisions → measure cost impact.



Combined effect: Batching + Caching + Cascading +

Quantization → 70–85% cost reduction so với naive deployment.

- Group 10 requests mỗi batch
- Throughput tăng 8x, cost/request giảm 85%
- vLLM continuous batching tự động
- Tune max_num_seqs theo latency SLO

- Redis cache cho identical prompts
- Hit rate 30–40% điển hình cho chatbot
- Semantic cache: embed prompt → similarity search
- TTL: 1h cho dynamic, 24h cho static prompts

- Small model (Llama-3-8B) xử lý 80% requests
- Escalate 20% complex queries → large model
- Router: classify difficulty bằng fast classifier
- Cost reduction: 60–70%

Mode	Tok/s	\$/M tok
FP16 (A10G)	1200	\$0.83
AWQ 4-bit	1800	\$0.55
Savings		34%

Spot cho inference: stateless inference servers phù hợp spot instances nếu có load balancer automatic failover.

- `torch.cuda.amp.autocast`: forward pass FP16
- GradScaler: tránh underflow khi backward
- Memory giảm ~30–40% → batch size lớn hơn
- Training time giảm 20–50% tùy model
- Accuracy gần như không đổi (<0.5% drop)

Metric	FP32	AMP
Time/epoch	1.0x	0.6–0.8x
Peak memory	100%	60–70%
Cost/run	\$1.00	\$0.60–0.80
Savings		20–40%

BF16 trên A100/H100: không cần GradScaler, stable hơn FP16

Lab demo: Train ResNet-18 CIFAR-10 FP32 vs AMP trên Kaggle GPU → so sánh time, memory, power, cost thực tế.

- team=ml-platform
- project=rag-service
- env=production
- cost-center=engineering
- Enforce tags bằng SCP/OPA policies

- ResourceQuota per namespace
- Team budget: max 4 GPUs, 100GB storage
- **Kubecost:** per-pod cost breakdown
- “RAG service \$45/day, Embedding \$12/day”

AWS Cost Explorer + Budgets: alert khi spending vượt \$1,000/day → immediate investigation. **Monthly FinOps review:** share dashboard → ownership drives optimization.

- Chạy training khi grid sạch nhất
- us-west-2 (Oregon hydro) 10x greener than us-east-1
- CodeCarbon library: track CO₂ per experiment
- Schedule heavy jobs off-peak hours

- Distilled models: Phi-3-mini vs GPT-4 — 100x smaller, 70% accuracy
- Chain-of-thought tăng cost 3–5x — chỉ dùng khi cần
- Green AI metric: CO₂ grams per 1000 inferences
- Track & report cùng performance metrics

Takeaway: Sustainable AI không chỉ “nice-to-have” — region selection + model selection = cost savings + carbon reduction cùng lúc.

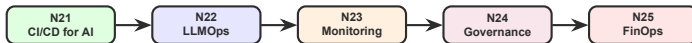
1. **Part 1–2:** Cluster monitoring + workload submission → đo utilization, phát hiện idle GPUs
2. **Part 3:** Spot bidding + preemption simulation → savings report (60–70% discount)
3. **Part 4:** Autoscaler policy tuning → observe scale-up/down decisions
4. **Part 5:** Cost tracker (OpenCost-like): waste report + optimization recommendations
5. **Part 6–7:** Visualization + end-to-end FinOps workflow
6. **Part 8:** Real GPU training FP32 vs AMP trên Kaggle → đo time, memory, cost thực tế

Mục tiêu: GPU FinOps Optimization Workshop

Deliverable: Lab notebook 8 parts (Docker mock cluster + Kaggle real GPU) + cost charts + Milestone 2 demo

Thời gian: 2.5h

Chương 5: Operations Layer Complete



- **Key insight:** operations cost thường = infrastructure cost sau 6 tháng production
- **FinOps quick wins:** schedule idle shutdown + quantize models + implement caching = 40–60% reduction
- **Invest early:** CI/CD + monitoring + governance + FinOps — trả nợ sớm, không phải trả lãi

Những ý chính cần nhớ trước khi sang bài tiếp theo

- 1 GPU cost anatomy: Compute 60% + hidden costs — audit waste trước khi optimize. Autoscaler giúp scale-down idle nodes tự động.
- 2 Spot instances tiết kiệm 60–70% — checkpoint mỗi 30 phút. Mixed Precision (AMP) giảm thêm 20–40% training cost.
- 3 Inference optimization: batching + caching + cascading + quantization = 70–85% cost reduction so với naive deployment.

Chương 6: Tổng Hợp — MCP/A2A Infrastructure

“Agent gọi agent — MCP server hosting, A2A protocol, agentic routing cho multi-agent systems”

- Hoàn thành Lab 25 + Milestone 2 demo
- Đọc trước: Anthropic MCP specification
- Đọc trước: Google A2A protocol overview

Hỏi & Đáp

Câu hỏi nào về GPU FinOps, spot instances, inference optimization, hay cost allocation?



Cảm ơn!

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